

DUPRE AGCM, AB, Canada

WMO: 714140

Lat: 54.367N

Lon: 110.868W

Elev: 556

StdP: 94.82

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-33.9	-30.9	-37.7	0.1	-33.6	-34.4	0.2	-30.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	27.8	17.4	26.1	16.7	24.5	16.0	19.4	25.0	18.2	23.8	17.1	22.5	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	13.2	22.6	16.0	12.2	20.8	15.0	11.4	19.5	57.7	25.0	53.6	23.9	50.3	22.6	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	DB	-39.1	30.6	2.5	1.2	-40.9	31.5	-42.3	32.2	-43.7	32.9	-45.5	33.7	(4)
(5)			WB	-39.2	21.9	2.5	1.5	-41.0	22.9	-42.4	23.8	-43.8	24.6	-45.6	25.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.9	-13.3	-13.7	-5.9	2.6	10.6	14.8	16.7	15.5	10.5	3.4	-6.1	-13.4
	DBStd	12.86	8.88	8.34	7.78	4.96	4.60	3.12	2.54	3.00	4.48	4.63	6.68	8.11
	HDD10.0	3628	722	663	493	226	49	5	0	2	47	211	483	727
	HDD18.3	6029	981	897	752	472	240	112	63	94	237	464	733	985
	CDD10.0	667	0	0	0	4	69	148	208	172	62	5	0	0
	CDD18.3	27	0	0	0	0	2	4	13	7	1	0	0	0
	CDH23.3	606	0	0	0	2	75	94	196	162	75	2	0	0
	CDH26.7	98	0	0	0	0	13	12	33	22	18	0	0	0
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	445	18	15	18	31	44	81	85	56	39	22	19	18
	PrecMax	585	36	29	38	92	88	178	169	148	92	62	57	43
	PrecMin	287	3	4	3	1	5	27	38	20	9	4	1	4
	PrecStd	73	9	7	10	19	23	38	30	27	23	13	13	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.9	6.6	12.2	22.3	28.4	28.2	29.5	28.8	29.3	21.8	10.9	5.4
		MCWB	2.7	2.9	5.8	10.3	14.1	16.3	19.4	18.3	16.8	11.9	4.5	1.7
	2%	DB	4.7	3.5	8.5	17.4	25.5	25.8	27.4	26.9	25.6	16.3	6.9	2.2
		MCWB	1.7	0.9	3.8	8.0	12.8	15.9	19.0	18.0	15.4	9.6	3.6	-0.3
	5%	DB	2.5	1.2	6.0	14.4	22.8	24.0	25.4	25.2	22.3	13.3	4.3	0.0
		MCWB	0.5	-0.8	2.4	6.4	11.8	15.2	17.8	17.5	14.0	7.7	2.1	-1.9
	10%	DB	0.1	-0.7	3.9	11.2	20.5	22.3	23.8	23.5	19.2	10.9	2.3	-2.3
		MCWB	-1.6	-2.0	1.1	5.0	10.6	14.2	16.8	16.2	12.5	6.3	0.6	-3.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.5	3.1	6.1	10.5	15.4	18.5	21.8	21.0	17.3	12.8	6.4	2.1
		MCDB	6.5	5.6	10.9	21.8	24.2	24.8	27.1	26.6	25.7	19.7	9.9	5.0
	2%	WB	1.9	1.0	3.9	8.2	13.9	17.3	19.9	19.3	16.0	10.2	4.0	-0.1
		MCDB	4.0	3.1	8.0	16.1	22.8	23.3	25.4	24.8	24.7	14.7	6.7	2.1
	5%	WB	0.4	-0.4	2.5	6.8	12.7	16.4	18.6	17.9	14.5	8.4	2.2	-1.7
		MCDB	2.5	1.1	5.5	13.9	20.5	21.8	24.1	23.4	21.4	12.6	3.9	-0.2
	10%	WB	-1.6	-2.1	1.3	5.4	11.5	15.3	17.5	16.9	13.1	6.8	0.8	-3.6
		MCDB	0.1	-0.6	3.5	10.4	18.7	20.2	22.5	22.3	18.3	10.5	2.2	-2.3
Mean Daily Temperature Range	5% DB	MDBR	10.7	11.8	11.3	11.1	13.8	11.8	12.2	13.2	13.2	10.2	8.5	9.2
		MCDBR	11.7	13.0	12.7	17.0	17.9	15.1	15.4	16.7	19.3	15.1	10.8	11.5
		MCWBR	9.3	10.6	8.5	9.0	7.6	6.8	8.0	8.6	9.6	8.9	7.7	9.5
	5% WB	MCDBR	11.6	12.3	11.7	15.7	16.0	12.9	13.8	15.1	18.0	14.4	9.9	10.6
		MCWBR	9.5	10.3	7.9	8.6	7.2	6.8	7.6	8.3	9.3	8.7	7.4	8.8
Clear-Sky Solar Irradiance	taub	0.252	0.274	0.283	0.331	0.347	0.364	0.362	0.369	0.311	0.279	0.251	0.227	
	taud	2.284	2.261	2.375	2.377	2.388	2.350	2.408	2.341	2.513	2.574	2.454	2.318	
	Ebn at Noon	720	822	907	898	899	883	878	847	868	831	737	688	
	Edh at Noon	55	81	91	105	110	116	108	108	80	60	47	43	
All-Sky Solar Radiation	RadAvg	0.87	1.91	3.44	4.66	5.69	5.83	5.92	4.83	3.39	1.89	0.88	0.61	
	RadStd	0.09	0.15	0.24	0.39	0.47	0.40	0.32	0.38	0.36	0.30	0.11	0.06	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page